

Tanzim Ikram Sheikh

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EDUCATION

American International University–Bangladesh (AIUB)

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering (CSE)

June 2021 – December 2025

Thesis: "Integrating Therapy and Technology: An Exploratory Case Study on the Development of Aids for Dementia"

CGPA: 3.82/4.00

RESEARCH INTERESTS

Human-Computer Interaction (HCI), Human-Centered AI, Human-Robot Interaction, Extended Reality (XR), Tangible Interaction, Multimodal and Gesture-Based Interfaces, User Experience Design, Smart Healthcare Systems, Assistive Technologies for Aging and Dementia, Human-Centered Design, Artificial Intelligence and Embedded Systems.

PUBLICATIONS

- **T. I. Sheikh**, M. R. Al Sultan, F. Ahmed, M. M. Iqra, and K. T. Hasan. Integrating Therapy and Technology: An Exploratory Case Study on the Development of Aids for Dementia in Bangladesh. ITAP 2026 (HCII), Lecture Notes in Computer Science (LNCS) 2026, 15903.
- S. M. Rohan, **T. I. Sheikh**, F. B. Zaman, M. Ahmed, M. S. Kabir, and K. T. Hasan. ReedMap: A Tangible Interaction System for Context-Aware and Division-Based Map Navigation. DUXU 2026 (HCII), Lecture Notes in Computer Science (LNCS) 2026, 15893.
- **T. I. Sheikh**, S. M. Rohan, N. Aziz, A. Amin, S. F. Mahbub, and K. T. Hasan. AirTune: A Tangible Interaction System for Gesture-Based and Contact-less Media Control. HCI 2026 (HCII), Lecture Notes in Computer Science (LNCS) 2026, 15899.
- K. Redwan, M. Ahmed, **T. I. Sheikh**, K. M. S. Hasan, M. F. A. Al Sohan, and K. T. Hasan. GlobeStick: Innovating Map Navigation Through Tactile Interaction and User-Centered Design. DUXU 2025 (HCII), Lecture Notes in Computer Science (LNCS) 2025, 15798.
- M. Ahmed, A. Sultana, P. B. Pew, S. Datto, **T. I. Sheikh**, K. Redwan, and M. F. A. Al Sohan. Melanoma Detection Using Augmented ResNet34 for High-Precision Dermoscopic Image Classification. ICCIES 2025, Lecture Notes in Networks and Systems (LNNS) 2025.

WORK EXPERIENCE

The Abroad Company

Remote

Product Experience Engineer

December 2025 – Present

• AbroadMates

- Designed the mentor landing page and streamlined the mentor signup flow, achieving a 100% mentor profile completion rate.
- Built a comprehensive mentor onboarding system, reducing drop-off and improving activation across the platform.
- Redesigned student and mentor dashboards with responsive data visualization and clearer information architecture, improving cross-device usability.
- Architected end-to-end user flows for an algorithmic mentor recommendation engine and a concurrent multi-mentor booking system with automated bundle discounts (e.g., 10% off for 5 mentors).
- Improved mentor discoverability by redesigning profile cards and detailed mentor pages.

• ApplicationMate

- Contributed to the foundation of a student-focused digital platform aimed at simplifying higher-education decision making through thoughtful interface design and workflow planning.
- Established the product's design direction by building a unified design system spanning onboarding, dashboards, and core workflows.
- Designed decision-support interfaces for country, subject, and program recommendations, enabling users to navigate complex study-abroad choices through clearer, more structured guidance.
- Designed the centralized student profile so applicants can enter data once and reuse it across applications.

Envolead Ltd.

Hybrid

Digital Innovation & Impact Intern

May 2025 – October 2025

- Built and deployed a full-stack company website, including landing page, admin dashboard, and CMS, using Laravel, Vue.js, and MySQL.
- Designed the EnvoYouth mobile application and established a consistent design system aligned with brand guidelines.
- Automated bulk personalized invitation letter generation and email distribution using Python, reducing manual outreach effort.

- Developed a real-time graphical interface for telemetry monitoring, data logging, and diagnostics of CanSat and flight computer systems.
- Built an ESP8266-based web server interface for the AERD Launch Box to support hybrid rocket fuel grain testing and monitoring.
- Collaborated across multidisciplinary engineering teams spanning aerospace experimentation, embedded systems, and interface development.

UX RESEARCH CASE STUDIES

Namjari App (Government Land Mutation App)

Team Lead, Group UX Case Study

Led a cross-functional team to redesign Bangladesh’s land mutation process into a guided, one-stop digital solution with unified payments and an AI chatbot, "Bhumisathi." Directed usability testing that reduced task completion time by 60% and raised task success rate from baseline to 100%, applying accessibility and trust-centered design principles for citizen users.

NID Wallet App (Government Digital ID App)

Team Lead, Group UX Case Study

Directed the redesign of the NID Wallet app into NID Sheba, a unified digital ID solution built around competitive analysis and a trust-centric onboarding flow with multilingual support. Led usability testing that improved task success from 60% to 100% and cut completion time by approximately 80%, and integrated biometric verification to strengthen security and user trust.

PROJECTS

- **AirCursor:** A dual-mode ESP32 wireless controller that uses an MPU6050 IMU and BLE HID to function as a gesture-controlled mouse and a hand-motion presentation remote, with tactile buttons for clicks, scrolling, slide navigation, and laser-pointer toggling.
- **MoodDial:** A tangible, emotion-aware interaction device that lets users express their emotional state via a physical rotary dial instead of a digital interface. Transmits mood data over WiFi to a backend AI model, which delivers supportive, mood-appropriate responses through Telegram, enabling low-effort, asynchronous emotional support.

TECHNICAL SKILLS

- **Design:** Figma, User Research, Usability Testing, Wireframing, Prototyping, Design Systems, User Flows, Information Architecture
- **Programming Languages:** Python, JavaScript/TypeScript, PHP, C/C++
- **Research & Data Tools:** NumPy, Pandas, Matplotlib
- **Web Frameworks:** Next.js, React, Node.js, Nest.js
- **Embedded Systems:** Arduino, ESP32/ESP8266, Sensor Integration, Circuit Design
- **Tools:** Git, Postman, Adobe Creative Suite, EasyEDA, Overleaf, VS Code, Jupyter Notebook, Google Colab

HONOURS AND AWARDS

1st Place (Joint Winner), Group UX Case Study Competition Ostad Limited	2025
1st Runner-Up, Logo Design Competition, AIUB CS Fest AIUB	2024
Leadership Award, English Olympiad Season 2 English Olympiad	2019

LANGUAGE PROFICIENCY

Academic IELTS

Overall: 7.5 | CEFR Level: C1

Listening: 8.5, Reading: 8.5, Writing: 6.5, Speaking: 6.5

EXTRA-CURRICULAR ACTIVITY

- Campus Ambassador, English Olympiad
- Senior Member, Connecting Youth for Change
- Participant, Inter University Line Following Robot Competitions

REFERENCES

Dr. Khandaker Tabin Hasan

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Department of Computer Science
Faculty of Science and Technology
American International University–Bangladesh
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Md. Mehzabul Hoque Nahid

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